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Chapter 4

Playing the Odds: Pot Odds and Implied Odds

The heart of poker is decision-making. Players who make better decisions than their opponents win; players who make worse decisions than their opponents lose. In Part I, we defined decisions with higher EV as "better" than their lower EV counterparts. In Parts II and III, we will examine the decision-making process as it applies to playing poker games and identify both techniques of analysis and methods of play that lead to making the best decisions at each juncture.

Part II deals with what we call ***exploitive play;*** this is play which seeks to maximize EV by taking the action at each decision point which has the highest EV in a particular situation, inclusive of whatever information is available about the opponent, such as his tendencies, tells, and so on. Virtually every player uses exploitive play in one form or another, and many players, even some of the strongest players in the world, view exploitive play as the most evolved form of poker.

Before we get into a discussion of exploitive play, we will introduce some terms and definitions. First, we have the concept of a ***game.*** In the poker world, we have different definitions of this term, and in Part III, we will be primarily working in the domain of ***game theory,*** the mathematical study of games. For now, we will define a game as containing the following elements:

- There are two or more players.
- At least one player has a choice of actions.
- The game has a set of outcomes for each player.
- The outcomes depend on the choices of actions by the players.

Normally in our poker discussions, there will be two or more players, and both players will have action choices. The set of outcomes for the game will be expressed in dollars won or lost.

Additionally, we call the "choice of action" a ***strategy. In*** game theory terms, a strategy is a complete specification of a player's actions choices at all possible paths the hand might follow. In poker, strategies are extremely difficult to specify, as we have what might be called a combinatorial explosion of paths. There are 1326 starting hands a player might hold. Then there are 19.600 different three-card flops that might hit, 47 different turns, and 46 different rivers. Even after factoring in some equivalences with regard to suit, we still have well over five million board/hand combinations to consider. Then we must specify how we will play-each of our hands on each street, how we will respond to raises, checks, bets, and so on.

This is basically impractical for any but the simplest toy games. As a result, we often use the term strategy a little more loosely in poker. Frequently, when we use this term we are referring to our expected play on this and perhaps one more street. The depth to which we specify the strategy is often tied to the convenience with which we can express it; simpler games and more static boards can often go deeper than more complex ones. It is normally just simpler to treat the game this way. We do, however, try to tie together the play on two or more streets as much as possible.

The concepts of Part I have particular meanings when we consider the play in terms of strategies. It is fairly meaningless to consider the expectation of a hand in a vacuum before the play begins, so instead we use the term "expectation of a hand" here to mean the expectation of a hand played with a given strategy against an opposing strategy. Likewise, the expectation of a hand distribution against a strategy is the weighted average expectation of the hands in that

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distribution against the opposing strategy, and so on.

Maximizing expectation against the opponent's strategy is the goal of exploitive play. If our opponent plays a strategy S, we define the ***maximally exploitive strategy*** to be the strategy (or one of the strategies) that has the highest expectation against S. When playing exploitively, it is often our goal to find the strategy that is maximally exploitive and then employ it. By doing this, we maximize our expectation. We begin with a simple toy game to illustrate the process of finding this strategy.

Example 4.1

Two players play headsup limit poker on the river. Player A has either the nuts (20% of the time) or a valueless (or ***dead)*** hand (80% of the time), and Player B has some hand of mediocre value - enough to beat dead hands, but which loses to the nuts. The pot is four big bets, and A is first. Let us first consider what will happen if A checks. B could bet, but A knows exactly when he has B beaten or not; hence he will raise B with nut hands and fold at least most of his bluffing hands. B cannot gain value by betting; so he will check. As a result, A will bet all of his nut hands. A might also bet some of his dead hands as a bluff: if B folds, A can gain the whole pot.

We'll call the % of total hands that A bluffs with ***x.*** A's selection of *x* is his strategy selection. B loses one bet for calling when A has a nut hand, and wins five bets (the four in the pot plus the one A bluffed) when A has a bluff. B's calling strategy only applies when A bets, so the probability values below are conditional on A betting. Using Equation 1.11, the expectation of B's hand if he calls is:

$\langle B, call \rangle = p(\text{A has nuts})(-1) + p(\text{A has a bluff})(+5)$
$\langle B, call \rangle = (0.2)(-1) + (5)x$
$\langle B, call \rangle = 5x - 0.2$

If B folds, his expectation is simply zero.

$\langle B, fold \rangle = 0$

We consider a few potential values for x:

Situation	x value	<B, call>	<B, fold>
A never bluffs	0	-0.2	0
A always bluffs	0.8	+3.8	0
A bluffs 5%	0.05	+.05	0
A bluffs 4%	.04	0	0

B should choose the strategy that has higher expectation in each of these cases. If A bluffs often B should call all the time. If A bluffs rarely, B should never call.

To determine how often A will bluff, B might use his knowledge of A's play, tells, or some other information, perhaps using Bayes' theorem (A might have lost the last hand and therefore has a higher *a priori* chance to be playing this hand overaggressively because of emotion, etc.).

We can also graph these two functions:

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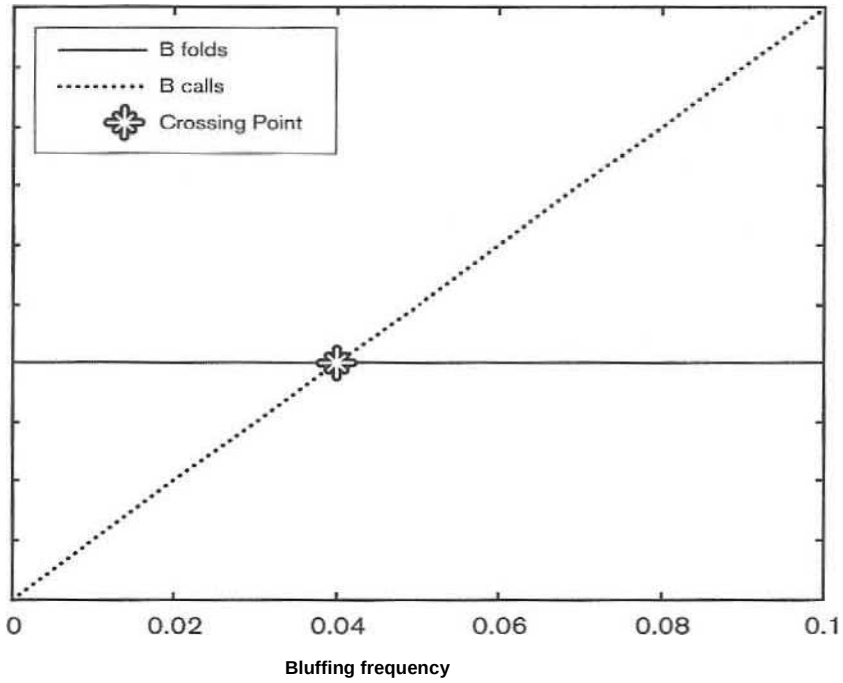

Figure 4.1, Game Equity for various B strategies

Figure 4.1 shows the linear functions that represent B's strategy options. We can see that B's maximally exploitive strategy involves choosing the strategy that has the higher value at the x-value that A is playing. So when x is below 0.04, (meaning that A is bluffing 4% of the time), B should simply fold; above that B should call. One tiling that is important to note about this is that exploitive play often involves shifting strategies rather drastically. If A changes his x-value from 0.039 to 0.041, that is, bluffing two thousandths of a percent more often, B changes his strategy from folding 100% of the time to calling 100% of the time.

Over the next several chapters, we will look, at some of the principles of exploitive play, including pot odds and implied odds, and then consider the play of some example hands. We - will consider situations where the cards are exposed but the play is non-trivial and then play a single hand against a distribution of hands. We focus primarily on the process of trying to find the maximally exploitive strategy in lieu of giving specific play advice on any given hand, and especially on the process of identifying specific weaknesses in the opponent's strategy. This last is particularly valuable as it is generally quite repeatable and easy to do at the table.

Pot Odds

None of the popular forms of poker are ***static games*** (where the value of hands does not change from street to street). Instead, one common element of all poker games played in casinos is the idea of the ***draw.*** When beginners are taught poker, they often learn that four cards to a flush and four cards to a straight are ***draws*** and that hands such as pairs, trips, and flushes and straights are ***made hands.*** This is a useful simplification, but we use "draw" to mean a variety of types of hands. Most often, we use "draw" to refer to hands whose value if the hand of poker ended immediately is not best among the hands remaining, but if certain cards (often called outs) come, they will improve to be best. However, in some cases, this can be misleading. For example, consider the following two hands on a flop of T♣ 9♣ 2♦ in holdem:

Hand A: Q♣ J♣
Hand B: A♠ 3♦

Hand A has more than a 70% chance of whining the hand despite his five-card poker hand being

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worse at this point. In this hand, it may seem a little strange to refer to the hand that is more than a 7 to 3 favorite as "the draw" while the other hand is the "made hand," because of the connotations we usually associate with these terms. However, we will consistently use the term "draw" throughout this book to mean the hand that at the moment has the worse five-card poker hand and needs to catch one of its outs, no matter how numerous they may be, in order to win. In contrast, we will use the term "favorite" to mean that hand that has more equity in the pot and "underdog" to mean the hand that has less. In the above example, the Q♣ J♣ hand is both the favorite and a draw.

One of the fundamental confrontations between types of hands m poker is between made hands and draws. This confrontation is particularly accentuated in limit poker, where the made hands arc unable to make bets large enough to force the draws to fold; instead, they simply extract value, while (he draws call because the expectation from their share of the pot is more than the amount they must call. However, all is not lost in big-bet poker for the draws. As we shall see later on, draws are able to take advantage of the structure of no-limit games to create a special type of situation that is extremely profitable by employing a play called the ***semi-bluff.***

Example 4.2

The game is $30-60 holdem. Player A has A♣ A♦. Player B has 9♥ 8♥. The board is K♥ 7♣ 3♠ 2♥. The pot is $400. Player A is first. How should the action go if both players know the full situation?

You can likely guess that the action goes A bets - B calls. It is valuable, however, to examine the underlying mathematics because it provides an excellent introduction to this type of analysis and to the concept of ***pot odds.***

If Player A checks, then Player B will certainly check behind. If B were to bet, he would immediately lose at a minimum $^3/_5$ of his bet (because he only wins the pot $^1/_5$of the time), plus additional EV if Player A were to raise. There will be no betting on the river, since both players will know who has won the hand. Rather, any bet made by the best hand will not be called on the river, so effectively the pot will be awarded to the best hand.

Since 35 of the remaining 44 cards give AA the win, we use Equation 1.11 to determine A's EV from checking to be:

$<A, check> = p(\text{A wins})) + (\text{pot size})$
$<A, check> = (^{35}/_{44})(400)$
$<A, check> = \$318.18$

Now let's consider B's options. Again, B will not raise, as he has just a $^1/_5$ chance of winning the pot (with 9 cards out of 44) and A will never fold. So B must choose between calling and folding.

$<B, call> = (p(\text{B wins}))\ (\text{new pot size}) - (\text{cost of a call})$
$<B, call> = (^9/_{44})\ (\$400+60+60) - \$60$

$<B, call> = (^9/_{44})\ (\$520) - \$60$
$<B, call> = \$46.36$
$<B, fold> = 0$

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Since B's EV from calling is higher than his EV from folding, he will call in response to from A. That makes A's EV from betting:

<A, bet> = (*p(A* wins)) (new pot size) - (cost of a bet)
<A, bet> = $(^{35}/_{44})$ ($520) - $60
<A, bet> = $353.64

Comparing A's equity from betting to A's equity from checking, we see that A gains a more than $35 by betting. This result indicates the main principle of made hand vs. confrontations: ***In made hand vs. draw situations, the made hand usually bets.***

There are some exceptions to this rule, as we shall see later when we consider some: complicated draw vs. made hand games. Now let us consider B's play. We found that equity from calling was greater than the zero equity from folding. We can solve an inequity to find all the equities at which B would call, letting *x* be B's chance of winning the pot:

(B's chance of winning the pot) (new pot size) - (cost of a call) > 0
520x - 60 > 0
$x > {}^{3}/_{26}$

So B ought to call if his probability of winning the hand is greater than $^{3}/_{26}$, which they are the given case. As we discussed in Part I, we might also say that the odds of B winning be at least 23 to 3. We often call this relationship B's ***pot odds.*** B should call if his odd winning the pot are shorter than his pot odds. Pot odds are a shorthand for straightfor equity calculations-the pot lays *X* to our bet of *Y*; our odds of winning are *W* to *Z,* hen choose a decision based on the relationship between those two.

This leads us to the second principle of hand vs. draw situations.

In made hand vs. draw situations, the draw usually calls if it has positive equity in the after calling and subtracting the amount of the call.
There are a number of exceptions to these rules, and by studying individual hands w examine the situations in which simply relying on pot odds as a guide is incorrect or income. However, pot odds calculations are a useful and practical way of approximating and some calculating outright whether it's correct to call with a draw, particularly at the table.

Pot Odds on Multiple Streets

Poker, of course, is a multi-street game, and as such, we often confront situations where it simply action on the current street that needs to be considered, but action on future streets as Consider another perfect-information situation (both players know their opponents' cards).

Example 4.3 ($75 Pot)

Is the flop of a $30-$60 holdem game. Player A holds A♦ K♦, and player B holds 8♣. The flop is A♣ Ks 4♣. There is just $75 in the pot (suppose that the AK limped preflop) the AK bets $30. B has just eight outs in this case (as the K♣ makes AK a full house). So an immediate pot odds standpoint, B should call if his chance of winning the pot is go than 30/105, or 28.5%. So what is B's chance of making his flush on the next street? It’s on the turn. This chance is only 17°/o, and so we can see that B's chance of making his is not high enough to justify calling.

Example 4.4 ($195 Pot)

What if the pot were larger, say $195? Now B has enough immediate odds to call with his flush

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draw on the flop. His EV is then $(^{8}/_{45})$($255) - $30 = $15.33. On the turn, if the draw misses, there are two cases. In one case, an ace or king will have hit on the turn and he will be unable to win the pot, or be ***drawing dead.*** This occurs $^{4}/_{45}$ of the time. The remaining $^{41}/_{45}$ of the time he will have $(^{8}/_{44})$ ($375) - $60 = $8.18 of positive expectation from calling.

His total expectation on the turn, again using equation 1.11, is:

<B, turn> = *p(A* or K) (<B, A or K hits) + *p*(other)(<B, other>)
<B, turn> = $(^{4}/_{41})$ (0) + $(^{41}/_{45})$ ($8.18)
<B, turn> = $7.45

So in this case, the play on the flop would go A bets - B calls. Then if the draw did not hit on the turn, A would bet again and B would be able to profitably call again (because calling is worth more than zero from folding).

Example 4.5 ($135 Pot)

There is also another pot size of interest. If the pot were just $135, B has enough equity to justify calling on the flop, but not on the turn.

<B, flop> = *p*(win on turn) ($135 + $60) - $30
<B, flop> = $^{8}/_{45}$ ($135 + $60) - $30
<B, flop> = $4.67
<B, turn> = *p*(win on river)($195 + $120) - $60
<B, turn> = $(^{8}/_{44})$ ($315) - $60
<B, turn> = -$2.73

(Here we ignore the case where A hits an ace or king on the turn and wins the hand immediately. Since B can't call when we ignore this case, his equity is worse when we include those hands, and hence he can't call if we include it.)

This poses a difficulty with our previous characterization of pot odds. We stated earlier, "B should call if his odds of winning the pat are shorter than his pot odds." But here B's chance of winning the pot by the end of the hand is:

p(B wins) = 1 - *p(B* loses)

By applying Equation 1.3:

p(B wins) = 1 - [*p*(B loses on turn)][*p*(B loses on river)]
p(B wins) = 1 - $(^{37}/_{45})$ $(^{36}/_{44})$
p(B wins) = 0.327 or 32.7%

Here we use the joint probability $p(A \cap B) = p(A)p(B)$ of B not whining the pot and subtract it from one to find the probability of him winning the pot. We could likewise calculate the chance of him winning on the turn and also on the river, and then subtract the chance that he wins on both streets. However, the single step above is more straightforward for this type of calculation.

If B has nearly a one-third chance of winning the pot then why can we not call in all of these cases, since in each of them B is getting longer odds from the pot than two to one? For example, in 4.3, the smallest pot of the three, B is getting ($75+$60) to $60, or 2.25 to 1. The answer to this question is that only situations where B wins on the next street count for his immediate pot

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odds.

Example 4.6 (Why giving yourself odds by raising doesn't work)

Consider the case we looked at previously where the pot is $135 on the flop. If B calls the flop, there is $195 in the pot on the turn. If the draw misses, then when A bets $60, B lacks the odds to call the bet on the turn. But what if B raises the flop instead? Now A will reraise, B will call, and B will have odds to draw on the turn as well. It turns out that this is a losing play for B compared to simply calling the flop and folding the turn. (If B hits one of his outs, he will bet and shut A out of the pot, so the full house redraw doesn't come into play.)

We can break B's expectation into two elements. One is his expectation if he wins on the turn, and the other is the expectation when the hand goes to the river. Let's call B winning on the turn card *Bf* and B not winning on the turn card *Bt*.

<B, raise flop> = <*Bf*, raise flop> + <*Bt*, raise flop>
<*Bf*, raise flop> = *p*(B wins on the turn) (pot if B raises and A reraises) - (cost of the raises)
<*Bf*, raise flop> = ($^{8}/_{45}$)($135 + 2($30+$30+$30)) - $90
<*Bf*, raise flop> = -$34
<*Bt*, raise flop> = *p*(B doesn't win on the turn)[*p*(B wins on the river)(pot after the turn bets) – (cost of the turn bet)]
<*Bt*, raise flop> = ($^{37}/_{45}$)[($^{8}/_{44}$)($315 + 2($60)) - $60]

<*Bt*, raise flop> = $15.70
<*B*, raise flop> = -$34 + $15.70
= -$18.30
<*B*, *call* flop> = *p*(*Bf*)($135+2($30))-$30
= $4.67

So raising the flop, even though it makes it such that B has odds to call the turn and draw to his flush again, has over $20 less expectation compared to simply calling. In real-life poker, both players don't have all the information. In fact, there will often be betting on future streets after the draw hits his hand. The draw can use this fact to profitably draw with hands that are not getting enough immediate pot odds.

Implied Odds

In the previous section, we discussed the concept of pot odds, which is a reasonable shorthand for doing equity calculations in made hand vs. draw situations. We assumed that both players knew the complete situation. This assumption, however, does not hold in real poker. After all, the purported drawing player could be bluffing outright, could have been drawing to a different draw (such as a straight or two pair draw), or could have had the best hand all along. As a result, the player with the made hand often has to call a bet or two after the draw comes in.

When we looked at pot odds previously, the draw never got any money after hitting his hand, since the formerly made hand simply folded to his bet once the draw came in. However, such a strategy in a game with concealed cards would be highly exploitable. As a result, the draw ran anticipate extracting some value when the draw comes in. The combination of immediate odds and expected value from later streets is called ***implied odds.***

Example 4.7

Consider the following example:

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Imagine the situation is just as in the game previously (Player A holds A♦ K♦ and player B holds 8♣ 7♣. The flop is A♣ K♠ 4♣.), but Player A does not know for sure that Player B is drawing. Instead of trying to arrive at some frequency with which Player A will call Player B's bets once the flush comes in, we will simply assume that A will call one bet from B on all remaining streets.

Earlier we solved for B's immediate pot odds given that the hands were exposed. However, in this game, B's implied odds are much greater. Assume that the pot is $135 before the flop. Now on the flop, A bets $30, We can categorize the future action into three cases:

Case 1: The turn card is a flush card.
In this case, B wins $195 in the pot from the flush, plus $240 from the turn and river (one bet from each player on each street). Subtracted from this is the $150 B puts into the pot himself. So B's overall value from this case is $285. This case occurs $^{8}/_{45}$ (or 17.8%) of the time, for a total EV contribution of $50.67.

Case 2: The turn card is not a flush card, but the river is.
In this case, B again wins $285, as one bet goes into the pot on both the turn and river. This joint probability (using Equation 1.3) occurs $(^{37}/_{45})$ $(^{8}/_{44})$ of the time, or about 14.9%. This results in a total EV contribution of $42.61.

Case 3: Neither card is a flush card.
This occurs the remaining 67.3% of the time. In this case, B calls the flop and turn but not the river, and loses those bets. So he loses $90. This results in an EV contribution of-$60.55.

Outcome	*p*(Outcome)	Value	Weighted EV
Turn flush	8/45	+$285	$50.67
River flush	(37/45)(8/44)	+$285	$42.61
No flush	l-[(8/45)+(37/45)(844)]	-$90	$60.55
Total	1		$32.43

Summing the weighted EV amounts, we End that this game has an expectation of about $32.73 for B. Contrast this to the expectation of just $4.67 in the same game if A never paid off B's flushes.

Effective Pot Size

Exploitive play in games such as no-limit holdem relies quite heavily on implied odds. It is frequently correct to take the opportunity to see a cheap flop with hands such as small or medium pairs or suited connectors or weak suited aces in order to attempt to flop a set or a strong flush draw. When doing this, however, it is important to take into account not just the amount that we will win when we do make a powerful hand, but the amount we might lose when that powerful hand loses anyway (as, for example, when the opponent flops a higher set than we). Also, we cannot assume that our opponents will simply pay us off for their entire stack, as some players do when they use the size of the stacks as a guide to what implied odds they have.

In the previous examples, the increased equity from implied odds allowed B to call at lower pot sizes and with less equity because it increases the size of what we might call "the effective pot." In Part III, we will see that "payoff amounts" are a very important part of games played between hand distributions containing various combinations of made hands and draws. In addition, these amounts are of critical importance when considering games where the draw is ***closed.*** By this, we mean the information of whether the draw has come in is not available to t both players. This

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is the case, for example, in seven-card stud, where the final card is dealt facedown. When this information is asymmetric, the player with the made hand is at a disadvantage because he must at least sometimes pay off the draw when the draw gets there. The reason for this is because if he does not, then the draw can exploit this by bluffing aggressively.

Bluffing

Bluffing is perhaps the most storied action in poker. One of the most exciting moments in any beginning poker player's career is when he first bluffs an opponent out of a pot. A lot of the drama that is captured in poker on television revolves around moments where a player makes I large bet and the other player must decide whether the player is bluffing or not when deciding to call. It is true in some sense, of course; bluffing is an integral part of poker and is the element of the game that differentiates it directly from games such as chess or backgammon.

A ***pure bluff*** is a bet with a hand that has no chance of winning the pot if called by the opponent's maximally exploitive strategy. A ***semi-bluff*** is a bet with a hand that might or — tight not be best at the moment, but which can improve substantially on later streets. Ordinarily, pure bluffs only occur on the last street of a hand, when no further improvement is possible, although it is possible for players to start bluffing with no hope of winning the pot on earlier streets. We often call this ***snowing.*** An example of an appropriate snow is when holding four deuces in deuce-to-seven lowball. Standing pat before the draw and bluffing is often a quite effective strategy because the opponent cannot hold a very strong hand (86543 b sing his best possible hand).

On the opposite end of the spectrum are ***value bets.*** These are bets that expect to have positive expectation even when called. In extreme cases, such as when a player holds the uncounterfeitable nuts on an early street, we can make pure value bets early in a hand. But like straight bluffs, it is frequency the case that with cards to come, most value bets can in fact be semi-bluffs (this depends on how strong the opponent's hand is).

However, between pure bluffs and pure value bets, there is a broad spectrum of bets, most of which are sometimes value bets and sometimes semi-bluffs. The clearest and easiest example of a pure semi-bluff with no value component is a weak flush draw, which might or might not have value if it pairs its cards, but has no chance of being called by a worse hand in terms of high card value. Still, this type of hand can frequently benefit from betting because the opponent may fold. We will return to semi-bluffing later in Part II. The next example, however, shows the exploitive power of bluffing.

Example 4.8

The game is $40-80 seven-card stud. The hands are exposed, except for the river card, which will be dealt face-down. We call situations such as these ***closed draws:*** that is, the information about whether the draw has come in is available to only one player.

Player A: 6♥ A♣ A♠ ***7♦ 9♦*** K♣
Player B: 7♠ 8♠ 9♠ K♠ 2♣ 4♣

As you can see, this game mirrors to some extent the previous game. Player A has a made hand, while Player B has a pure flush draw with no secondary outs. The pot is $655. The river card is dealt.

In this case, Player A is done betting for this hand. Either B has made his flush or he has not, and B will never call A's bet unless B beats A. Hence, A should simply check and then attempt to play accurately if B bets. If B checks, of course, A should expect to win the pot all the time. In

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fact, A should expect to win the pot quite frequently here, far more than 50°/o of the time. However, he should still not value bet because he only loses expectation by doing so. His expectation in the large pot is still intact whether he bets or not. In this case, also, A's value bet, no matter what card he catches, will never extract value from B's hand, because B can't make a hand better than the open aces A has.

So A checks. Now B must decide what to do. Clearly, he should value bet his flushes. There are 40 cards left in the deck, and eight flush cards. So B will make a flush $^1/_5$(or 20%) of the time. In addition to value betting his flushes, B might also want to bluff sometimes when he misses the flush. If he does this and A folds, he'll win a pot with more than eight bets in it with the worst hand.

If B bets, then A has to decide between calling and folding. This decision is actually affected by whether or not A caught a spade on the river, because if he did it is more likely that B is bluffing. But we'll neglect this for now; assume that A decides to play blind on the river, for our convenience.

This leaves us with two unknowns:

A's calling frequency: that is, given that B bets, how often A calls. Call this value ***x.***
B's bluffing frequency: that is, what percentage of his total hands B bluffs with. Call this value *y*.

Then, from Equation 1.11, we have the following equations:

<A, call> = *p(B* has a flush) (lose one bet) + *p*(B is bluffing) (pot + one bet)
<A, call>= (0.2)(-$80) + (y)($655 + 80)
<A, call> = $735y - $16
<B, bluff> = *p(A* calls) (lose one bet) + *p(A* folds) (pot)
<B, bluff> = x(-$80) + (1 - x)($655)
<B, bluff> = $655 - $735x

So the value of calling for A is dependent on how often B will bluff; the value of calling for B is dependent on how often A will call.

For A, we can solve an inequality to see when calling will have positive value:

$735y - $16>0
y > ~ 2.2%

This means that if B bluffs more than 2.2% of the time, A should call all the time, because he will have positive expectation by doing so. Also, we can see that at precisely this bluffing frequency, A's calling will have EV 0. This is a critical concept that we will revisit again and again in this book. At this frequency, A is ***indifferent*** to calling and folding-it doesn't matter what he does.

For B, we can solve an analogous inequality to see when bluffing will have positive value.

$655 - $735.x > 0
x< ~ 89.1%

This means that if A calls less than 89.1 % of the time, then B should bluff all the time, because he will have positive expectation for doing so. We can again see that at precisely this frequency,

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B's bluffs will have EV 0. That is, B is indifferent to bluffing or checking his non-flushes.

It might happen that Player A is a "skeptic ," who thinks that Player B is a habitual bluffer and will therefore call all the time. If he does this, his expectation from calling is as above, about $7.35 for every percent of dead hands that B bluffs (above the critical value). B's best response no A's skepticism is to simply stop bluffing. Or perhaps Player A is a "believer." who thinks that Player B wouldn't dare bet without having made a Bush. So he folds quite often. Then Player B can simply bluff all his non-flushes, gaining the entire pot each time that Player A folds. Of course, Player B loses out on value when he bets his flushes, but the eight-bet pots be wins more than offset that.

Exploitive Strategies

The responses of Player A and Player B to their opponents' different bluffing and calling frequencies are prototypical exploitive plays. Both players simply maximize their equity against their opponent's strategy. We call these strategies ***exploitive*** because by performing these equity calculations and employing these strategies, these players identify the weaknesses in their opponent's strategy and exploit them by taking the actions that maximize EV against those weaknesses.

However, equity calculations can be difficult to do at the table, particularly when there are many additional factors, such as those that occur in the real world. In this book, we often present toy games as conceptual aids to the processes we describe. As we examine more complicated situations, where the play is distributions of hands against each other and complex strategies are employed by both sides, sometimes doing straightforward equity calculations is beyond a human player's reach hi the time available at the table.

Nevertheless, the identification and exploitation of weaknesses in our opponents' strategies leads us to plays that increase EV against the strategy. And so if our ultimate goal is to maximize the results of equity calculation, identifying and exploiting weaknesses is an effective proxy for expected value calculation and is often more practical at the table.

Returning to Example 4.8, we presented this game as a sort of guessing game between A and B. where each attempts to guess the strategy that the other will employ and respond to that **by** moving their strategy dramatically in the other direction. However, we can briefly foreshadow things to come in Part III here. The indifference points we identified (~2.2% bluffs for B and 89.1% calls for A) are special values: they are the points at which each player's opponent cannot exploit the strategy any further, even if they know the strategy. If A and B each play these specific strategies, then neither player can improve his EV by unilaterally changing his action frequencies. These strategies are called ***optimal.*** Finding optimal strategies and playing optimally is the subject of Part III.

Strong exploitive play is essentially a two-part process. The first step is to gather information about the situation. This can include Inferring the opponent's hand or hand distribution from the action or identifying situations in which the opponent plays poorly or in a manner that is easily exploitable. The second step is deciding on action in light of the information gathered in the first step. This is often simply taking the exploitive action called for based on the weakness identified in the first step. The second step is often simpler than the first, but this is not always true, as we shall see. We will consider these two steps of the process in turn, with our eye always turned to maximizing EV through the process.

Key Concepts

- Exploitive play is the process of maximizing expectation against the opponent's hands

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and strategy. In practice, this often amounts to identifying weaknesses in the opponent's strategy and exploiting them because detailed calculations of expected value are too difficult at the table.

- Pot odds provide a metric for us when considering whether to call with a draw or not; if the size of the pot relative to our chance of winning is large enough, calling is correct; otherwise folding is often indicated.
- Pot odds across multiple streets must be treated together-sometimes if we lack pot odds on a particular street, we can still calf because the total cost of calling across all streets is low enough to make the entire play profitable. The focus must be on the total expectation of the strategy rather than the play on any particular street.
- Implied odds involve additional betting that takes place after a draw is complete. Figuring implied odds into calling or folding decisions can help us to maximize EV.
- Two principles of hand vs. draw situations:
  1) In made hand vs. draw situations, the made hand usually bets.
  2) In made hand vs. draw situations, the draw usually calls if it has positive equity in the pot after calling and subtracting the amount of the call.
- A pure bluff is a bet with a hand that has no chance of winning the pot if called by the opponent's maximally exploitive strategy. A semi-bluff is a bet with a hand that might or might not be best at the moment, but which can improve substantially on later streets. Value bets are bets that expect to have positive expectation even when called.

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